

# Module 8: Multi-Party Computation (MPC)

## Threshold Cryptography for Secure Signing

# What is MPC?

## Multi-Party Computation

- **Distributed Cryptography:** Multiple parties compute together without revealing secrets
- **Threshold Security:**  $t$ -of- $n$  scheme (need  $t$  shares out of  $n$  total to sign)
- **Secure Signing:** Sign transactions without exposing private keys

## Use Cases

- **Multi-sig Wallets:** Distributed control over funds
- **Secure Custody:** Institutional-grade key management
- **Decentralized Signing:** No single point of failure

# Threshold Schemes

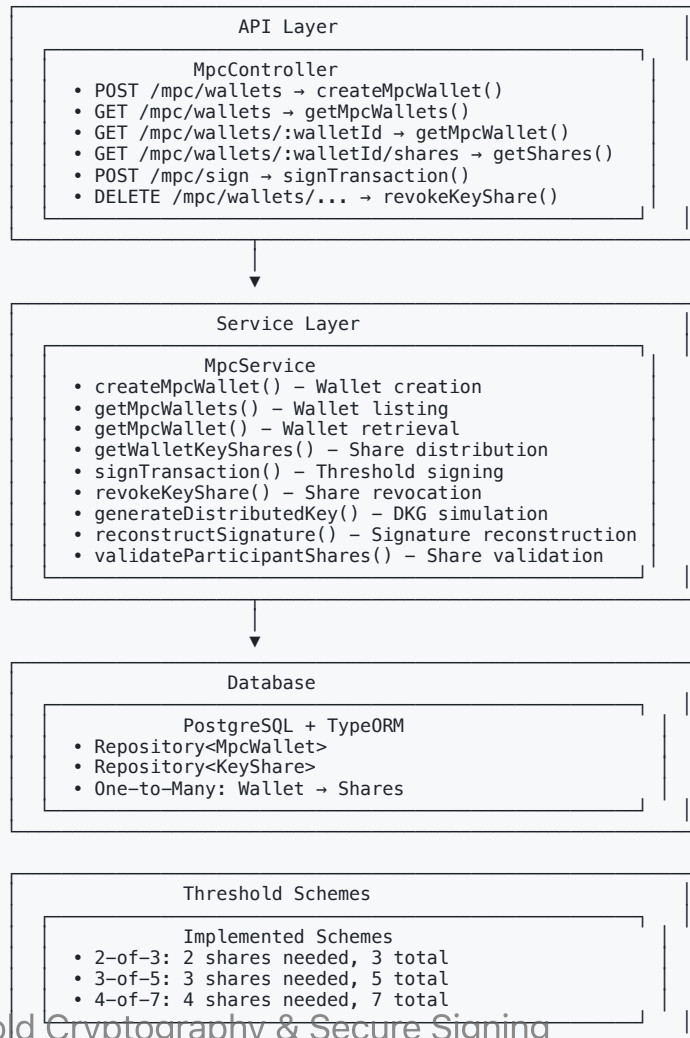
## Common Configurations

- **2-of-3**: 2 shares needed, 3 total shares
- **3-of-5**: 3 shares needed, 5 total shares
- **4-of-7**: 4 shares needed, 7 total shares

## Security Properties

- **t shares**: Minimum required for signing
- **n shares**: Total shares distributed
- **No single point**: No individual can sign alone
- **Flexible recovery**: Replace compromised shares

# Architecture Overview



## Threshold Signing

```
// Sign transaction with 2 shares (for 2-of-3 wallet)
const signature = await mpcService.signTransaction({
  walletId: wallet.id,
  transaction: serializedTransaction,
  shares: [
    { participantId: "alice", shareIndex: 0, encryptedShare: aliceShare },
    { participantId: "bob", shareIndex: 1, encryptedShare: bobShare }
  ]
});

// Use signature in Solana transaction
const tx = Transaction.from(Buffer.from(serializedTransaction));
tx.addSignature(wallet.publicKey, signature);
```

# Error Handling

## Common Errors

- **BadRequestException**: Invalid parameters or insufficient shares
- **NotFoundException**: Wallet or share not found
- **UnauthorizedException**: Invalid participant authentication
- **ForbiddenException**: Threshold not met for signing

## Validation Checks

- Participant public key verification
- Share encryption validation
- Threshold requirement enforcement
- **Wallet status verification**

## Use Cases & Benefits

### Institutional Custody

- **Secure Key Management:** No single employee can access funds
- **Regulatory Compliance:** Audit trails and access controls
- **Business Continuity:** Share recovery for lost access

### Decentralized Applications

- **DAO Treasury:** Multi-sig control for community funds
- **DeFi Protocols:** Secure oracle or bridge operations
- **NFT Marketplaces:** Escrow and royalty management

# Performance Considerations

## Optimization Strategies

- **Share Caching:** Cache decrypted shares for session
- **Batch Signing:** Process multiple transactions together
- **Connection Pooling:** Efficient database connections
- **Async Processing:** Non-blocking cryptographic operations

## Scalability

- **Horizontal Scaling:** Multiple MPC service instances
- **Load Balancing:** Distribute signing requests
- **Rate Limiting:** Prevent abuse and DoS attacks



# Security Best Practices

## Operational Security

- **Secure Share Storage:** Encrypted cold storage for shares
- **Access Logging:** Complete audit trail of all operations
- **Regular Rotation:** Periodic key share rotation
- **Backup Procedures:** Secure backup and recovery processes

## Cryptographic Security

- **Key Share Encryption:** Strong encryption for stored shares
- **Secure Communication:** TLS for all API communications
- **Participant Verification:** Multi-factor authentication
- **Compromise Response:** Immediate share revocation procedures

# Integration with Solana

## Transaction Signing

- **Ed25519 Compatibility:** Works with Solana's signature scheme
- **Transaction Serialization:** Standard Solana transaction format
- **Fee Management:** Integrated with fee estimation services

## Wallet Abstraction

- **Unified Interface:** Same API as regular wallets
- **Transparent Operation:** Applications don't need MPC awareness
- **Fallback Support:** Regular keypair signing as backup

## Next Steps

### Module 8 Implementation Tasks

-  Threshold signature generation
-  Distributed key generation
-  Key share management
-  Signature reconstruction
-  MPC wallet creation
-  Secure key distribution
-  MPC transaction signing
-  Recovery mechanisms

### Related Modules

Threshold Cryptography & Secure Signing

- **Module 7: Signing & Cryptography**

# Thank You!

## Questions?

**MPC enables secure, distributed control over blockchain assets!**

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